

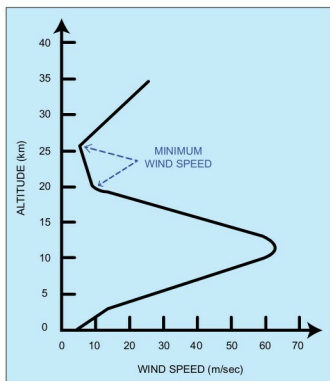


Fig. 2: Wind velocity with respect to the altitude

covering another zone. HAP acts as a hub for communication. Altitude of 18-25 km is chosen because the average wind speed at this height is minimum and the coverage of the antennae can be a footprint of up to 80km diameter with HAP altitude of 20 km, resulting in a cellular service for a large number of users over a wide area. The variation of wind speed with altitude is shown in Fig. 2.

Propagation and losses

Line-of-sight (LOS) path is required for communication through HAP. Signals at this frequency band are attenuated by rain. Polarisation diversity (where signals with different polarisations are used) may be acceptable in faded environments for good-quality signal reception.

In HAP communication, signal delay is negligible compared to direct satellite communication. Several studies verify that propagation path loss on the HAP link increases with the square of the distance (d^2) instead of d^4 as in terrestrial systems. The path loss in dB may be obtained using the relationship:

$$L = 32.4 + 10 \log f^2 + 10 \log d^2$$

where ' f ' is the frequency in MHz and ' d ' is the distance between the HAP

antenna and the user in kilometres. Here the curvature of the earth is neglected for a coverage diameter smaller than 100 km. This behaviour of the path loss is shown in Fig. 3.

If HAP is at an altitude of 20 km, the free-space loss at 1800MHz frequency band may be 120-130 dB. To handle the increased losses due to multipath fading in non-line-of-sight (NLOS) environment, automatic repeat request (ARQ) may be introduced. In ARQ scheme, after reception of erratic information from the transmitter, the receiver sends

a request to the transmitter through a feedback path to repeat the transmission again.

Presence of raindrops can severely degrade the reliability and performance of communication links at frequencies above 10 GHz. The attenuation due to rain can be expressed as:

$$A = aR^b$$

where ' A ' is the attenuation (in dB/km), ' R ' is the rain rate (in mm/hour), and ' a ' and ' b ' are factors depending on the rain drop size and frequency.

Co-channel interference and adjacent channel interference are the other important factors which may cause signal losses at the receiver antenna in HAP communication. Co-channel interference may increase due to cross-polarisation losses at the receiver antenna.

Requirement of antennae

Various types of antennae are used for broadband wireless communications, many of which are omnidirectional. In broadband wireless communication using HAP, directional antennae are required. Normal horn antennae, multibeam horn antennae or digitally-controlled array antennae may also be used for HAP. Very small aperture

AN ISO 9001-2008 Company

NSIC - CRISL
SE 28

Reliability in everything we do!



16/20 BATTERIES
SMPS CHARGERS

DIN RAIL
MOUNTING
SMPS



OVER VOLTAGE
PROTECTOR / OVCD



AC - DC / DC-DC CONVERTERS



SMPS ADAPTER / E-BIKE CHARGER



VOLTAGE &
CURRENT
PROTECTION
RELAY



AVR



BATTERY BACKUP UNIT

• Can withstand upto 440 AC
• Custom Builds SMPS / Battery Charger
• Universal IP: 90V-300V AC • Other Power Products



**NUTECK
PRODUCTS**
complete power solutions

115/5, Ramtek Industrial Area, Hadapsar, Pune 411 013.
Tel.: +91 20 26811161/62, Fax: +91 20 26811163
Email: info@nuteckproducts.com
Web: www.nuteckproducts.com

Contact: 09552545012

Dealers Enquiry Solicited